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M-406 SAMPLING RACK SYSTEM

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M 406 SAMPLING RACK SYSTEM

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with part 4, Subsection 1.2. “Summary of Work”.

The Sampling Rack System shall be provided for each block. The Bidder shall supply a complete water & steam sampling and analysis system to monitor the chemistry of the water and steam cycles of the HRSG turbine generator and associated equipment in order to control the chemical additives to the water cycles to ensure correct water and steam quality.

The Sampling Rack System for each block shall include the major components with accessories but not be limited to the followings :

- (1) Sampling rack
- (2) Analyzers, indicators and recorders
- (3) Sample controllers, sample coolers
- (4) Sampling piping with valves and supports
- (5) Special tools and standard tool sets
- (6) Cables
- (7) Anchor bolts, nuts, sleeves, etc.
- (8) Painting

Sampling rack shall be housed within an air conditioned enclosure.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

3. DESIGN AND OPERATING CONDITIONS

The sampling rack system shall be installed to supervise the quality of plant make up water, condensate, feed water, boiler water and steam, and shall

maintain the quality of the above liquids & steam in the adequate values so as to ensure safe plant operation.

The sampling rack shall be of open frame steel and self-supporting type. The sampling rack shall include a stainless steel sink.

Sample coolers shall be provided so that temperature of sampling water is in the range of 30°C to 42°C for supply of each sampling line.

Cooling water for sampling cooler shall be supplied from closed cooling water system, and returned to closed cooling water system after cooling of sample cooler.

Flow sight glasses shall be installed at the outlet of each sample cooler.

The Bidder shall submit the lists of the sample points to the Owner for review and approval.

4. DETAILED REQUIREMENTS

4.1 General

All components shall be assembled and provided with the necessary fixing for mounting on the foundations.

All sample points, devices and valves shall be furnished with suitable permanent name plates.

The temperature control equipment of unit sampler type or equivalent shall be provided as necessary in order to maintain the constant temperature range of the sample for the pH cells, carbohydrazide analyzer dissolved oxygen meter and silica meter.

The Bidder shall provide the special tools and standard tool sets for the sampling rack system.

4.2 Sampling Rack

Sampling rack shall contain all sample conditioning components, such as piping, tubing, pressure reducing devices, needle and flow control valves, shut off valves, unit-sampler, heat exchangers, resin column, analyzing cells, and thermometers.

The sampling rack shall also include a stainless steel sink.

All samples shall be installed on the sampling rack but not be limited to the following :

- Each pressure Steam, Saturated
- Each pressure Steam, Superheated
- Each pressure drum, water
- Each pressure Feed water
- Deaerator outlet
- Condensate, discharge of main condensate pump
- Closed cooling water system
- Make-up water

Sampling rack should be open, self supporting steel frame type, one side mounting.

Analyzers, indicator and junction boxes shall be provided on the sampling rack.

4.3 Piping and Valves

Drain valves and pipings shall be installed at the upstream of sampling cooler to enable flush out of sampling piping before initial operation.

Drain piping shall be collected to single header and discharged to the drain funnel.

All valves in sampling rack shall be of stainless steel.

All piping and tubing shall be of stainless steel, and the tube connections shall be of stainless steel Parker “Triplelock” connectors or equivalent.

Each sampling piping between the detecting (take out) point and sampling rack shall be of stainless steel.

The piping shall be in conformity with Part 4, Technical Specification M-001.

The Valves shall be in accordance with Part 4, Technical Specifications M-004 and M-005.

5. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct testing in accordance with Part 4, Subsection 2.6 Testing and Guarantees and shall guarantee the following system guarantees :

Bidder shall furnish all test equipment or tools required for normal maintenance, testing and calibration.

The Bidder shall prove that the equipment performs as specified. If any performance is not to the Owner's Satisfaction, the Bidder shall correct it and such correction or modification shall be made at no change in the contract price.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality Assurance/Control shall conform to the requirements of Part 4, Subsection 2.7.

The quality control procedures so specified in Part 4, Subsection 2.7 include, but not limited to, the following :

- System check out.
- System Calibration.

7. INSTRUMENTATION AND CONTROLS

The Bidder shall furnish and install complete instrumentation and control systems and devices for sampling rack system including but not limited to those listed and described in this Technical Specification.

The following analyzers, sensors for remote monitoring shall be installed in the sampling rack.

These signals shall be connected to BOP DCS and communicated to Plant Information Management System.

- pH analyzers

Each pressure superheated steam

Each pressure feed water (Economizer inlet)

Deaerator outlet feed water

- Each pressure drum water
- Condensate extraction pump discharge
- Make up water
- **Conductivity analyzers**
 - Make up water
 - Condensate (Deaerator outlet)
 - Each pressure feedwater (Economizer inlet)
 - Each pressure drum water
 - Each pressure superheated steam
 - Condensate (Condensate pump discharge)
 - Cation exchange (before and after)
 - Closed Cooling water
- **Dissolved Oxygen analyzers**
 - Each pressure economizer inlet or
 - Condensate water (Deaerator outlet)
- **Residual carbonylhydrazide analyzers**
 - Each pressure feedwater (Economizer inlet)
- **Silica analyzers**
 - HRSG Drum Water

The Bidder shall select the analyzer based on easy operation and maintenance.

The pH, conductivity, dissolved oxygen and silica analyses shall be specially designed for use on high purity water of low electrical conductivity. Particular attention shall be given to the calibration facilities included in the analyzer. The conductivity and pH recorder shall have alarm sets for each sample.

The analyzer shall be provided with control equipments to keep the constant temperature of sample water.

The Bidder shall furnish the power cables, control cables, cable ducts, trays, conduits and grounding wires for the sample conditioning system.

The Bidder shall be responsible for design, supply, installation and termination of all interconnecting field cables.

For detailed description of instrumentation and control and the description of local mounted instrumentation rack, refer to Part 4, Technical Specifications I-001 and I-003.

8. ELECTRICAL SPECIFICATIONS

The Bidder shall furnish and install all electrical equipment and accessories required for the sampling rack system.

All electric wiring, and cabling integral to the system shall be factory installed, and shall have convenient terminal points for external connections.

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